Hospital Chatbot for Color-Coded Clinical Pathways

A Three-Layered Architecture

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March 19, 2026

Presentation Outline

The Triage Bottleneck

The Clinical Context:

- Hospitals in LMIC's(Lower-to-Middle Income Countries) face a crisis of overcrowding, leading to dangerous bottlenecks across all levels of patient care.
- A major structural driver is the influx of non-urgent patients occupying critical resources, making **accurate, hospital-wide triage** an absolute necessity to prevent fatal delays.

The Solution:

- A digital triage chatbot that acts as the patient's first point of contact.
- It seamlessly evaluates symptoms to **filter and route patients to their respective hospital departments** strictly based on the calculated severity of their illness.

The Three-Layered Architecture

To safely evaluate patients in a pre-hospital setting, the system utilizes three interconnected computational layers:

1. **Natural Language Processing (NLP) Layer:** The Listener & Data Extractor.
2. **Deterministic Algebraic Layer:** The Core Mathematical Safety Net (TEWS).
3. **Bayesian Probabilistic Layer:** The Predictive Detective & Clinical Override.

This three layered approach ensures the system is clinically safer, more robust to missing data, and highly responsive to natural human communication.

Layer 1: NLP in Healthcare (History & Relevance)

What is Natural Language Processing (NLP)?

- NLP is a branch of Artificial Intelligence that enables computers to understand, interpret, and manipulate human language.
- **Relevance in Health:** It bridges the critical gap between unstructured patient narratives and structured clinical data required for medical algorithms.

Historical Context:

- NLP evolved from early 1950s machine-translation systems to logic-based algorithms in the 1980s.[1]
- Historically, clinical NLP relied on rigid rule-based systems that struggled with human nuance. Today, modern *multimodal models* combining unstructured free-text with structured data consistently outperform traditional triage systems.[2]

[1] K. Sparck Jones, "Natural Language Processing: A Historical Review," *Linguistica Computazionale* (1994): 1-16.

[2] J. Stewart et al., "Applications of NLP at ED triage: A narrative review," *PLOS ONE* 18, no. 12 (2023).

Layer 1: The Pre-Hospital Data Problem

Relevance to this Project:

- Patients do not speak in standardized medical ontologies. A patient will say *"my chest is tight,"* not *"I am experiencing respiratory distress."*
- Furthermore, expecting a patient to input an accurate heart rate or blood pressure reading is an unrealistic expectation in a pre-hospital SMS/Web-based context in LMICs.

The AI Solution:

- We utilize a **Deep Attention NLP Model** to extract symptoms from natural dialogue, ensuring the system can accurately triage patients even when objective vital signs are missing or unmeasurable by the patient.

Layer 1: The NLP Pipeline (Text to Clinical Meaning)

The system converts raw patient text into computable variables through a strict mathematical pipeline:[3]

1. **Tokenization:** The raw text is broken down into individual, computable units called tokens (words or sub-words).
2. **Named Entity Recognition (NER):** The system classifies medically meaningful tokens (e.g., distinguishing a "Symptom" from a "Body Part").
3. **Contextual Embedding:** Tokens are mapped to dense vector spaces, allowing the AI to understand semantic relationships (e.g., knowing "hot" relates to "fever").
4. **Attention Scoring:** A softmax attention mechanism assigns mathematical weights to specific words based on clinical severity.
5. **Discriminator Detection:** High-weight tokens trigger specific SATS Clinical Discriminators to drive the final triage decision.

[3] D. Gligorijevic et al., "Deep Attention Model for Triage of Emergency Department Patients," *arXiv preprint* (2018): 1-5.

Layer 1: Pipeline Execution in the Ghanaian Context

Scenario: A patient in Kumasi describes their condition to the chatbot in Twi:

"Me ho yɛ me hye, me koko yɛ me ya, na me home te."
(Translation: I have a fever, my chest hurts, and I am short of breath.)

How the Pipeline Processes This:

- **Tokenization:** ["I", "have", "a", "fever", "chest", "pain", "and", "me", "home", "te"]
- **NER & Translation:** Recognizes "fever" (Condition), "chest pain" (Severe Symptom), and maps local colloquialisms ("me home te") to "Shortness of Breath".
- **Attention Scoring:** The model assigns massive mathematical weight to **[chest pain]** and **[home te]**, largely ignoring "I" and "and".
- **Discriminator Classification:** The system flags SATS Discriminators for suspected cardiac/respiratory distress, triggering an immediate **Red Pathway** override.

Layer 2: The Deterministic Model (TEWS)

The Objective Data Gap & Clinical Workflow:

- NLP perfectly extracts *subjective* symptoms from the patient's narrative, but calculating an accurate Triage Early Warning Score (TEWS) requires *objective* vital signs (v_k).
- **Data Acquisition:** Patients cannot reliably self-report vitals. Therefore, these hard metrics are exclusively measured and inputted into the system by the **triage provider** upon arrival.

The Core Equation

Once inputted by the provider, these vitals pass through a rigid **Deterministic Algebraic Model**:

$$TEWS = \sum_{k=1}^{n} w_k \cdot f_k(v_k)$$

Layer 2: Piecewise Function & Color Mapping

Example: Piecewise Function for Heart Rate ($k = 1$)

- The function mathematically evaluates the provider-inputted vital signs against strict clinical thresholds, completely removing human calculation errors:

$$f_1(v_1) = \begin{cases} 3 & \text{if } v_1 \geq 130 \text{ (Extreme Tachycardia)} \\ 2 & \text{if } 111 \leq v_1 \leq 129 \\ 0 & \text{if } 51 \leq v_1 \leq 100 \text{ (Normal Baseline)} \\ 2 & \text{if } v_1 \leq 40 \text{ (Severe Bradycardia)} \end{cases}$$

Algorithmic Color Mapping:

- **TEWS 7+** $\rightarrow$ **Red** (Immediate Resuscitation)
- **TEWS 5–6** $\rightarrow$ **Orange** ($<$ 10 mins)
- **TEWS 3–4** $\rightarrow$ **Yellow** ($<$ 60 mins)
- **TEWS 0–2** $\rightarrow$ **Green** (Divert to Polyclinic / Non-Urgent)

Layer 3: The Bayesian Probabilistic Model

The Flaw in Deterministic Logic: If an active heart attack patient has temporarily normal resting vitals, the Deterministic TEWS math will dangerously score them as **Green (0)**.

The Mathematical Solution (Bayes' Theorem): To calculate the hidden probability of a severe disease (D) based on the NLP-extracted symptoms (S), the system uses a Bayesian Belief Network:

The Bayesian Equation

$$P(D \mid S) = \frac{P(S \mid D) \cdot P(D)}{P(S)}$$

- $P(D \mid S)$: **Posterior** (The probability the patient has the severe disease given their symptoms).
- $P(S \mid D)$: **Likelihood** (The probability these exact symptoms occur in patients who actually have the disease).
- $P(D)$: **Prior** (The baseline prevalence of the disease in the hospital's population).

Layer 3: Execution of the Clinical Override

Real-World Execution Scenario:

- **User Input:** *"I am 70, freezing cold, confused, and haven't peed all day."*
- **Deterministic Score:** Vitals measured as normal $\rightarrow$ **Green Pathway**.

The Bayesian Calculation ($D =$ Septic Shock, $S =$ Confusion, Anuria, Cold):

- The network retrieves the historical Likelihood $P(S \mid D)$ and Prior $P(D)$ for Sepsis.
- The math evaluates to a high Posterior Probability: $P(SepticShock \mid S) = \mathbf{0.92}$ (**92%**).

System Action (SATS Discriminator):

- Because the probability of a critical condition is exceptionally high, the Bayesian layer triggers a **Clinical Override**, bypassing the TEWS score and instantly upgrading the patient from Green to **Red**.

System Safety: Under-Triage vs. Over-Triage

Minimizing Under-Triage (Risk: Near-Zero)

- Under-triage is a fatal clinical error. Our hybrid architecture relies on *double-redundancy*. For a critical patient to be missed, **both** the Deterministic and Bayesian engines must fail simultaneously.
- Recent validations of similar 3-tier triage tools report under-triage rates as low as **2.7%** (well below the global safety threshold of 10%).[4]

Accepting Over-Triage (Operational Trade-off)

- The Bayesian model's hyper-vigilance increases over-triage (typically **30%-48%**). However, over-triage is an operational inefficiency, not a clinical danger, and is easily absorbed by successfully diverting the "Green Tag" masses.

[4] R. Mitchell et al., "Validation of the Interagency Integrated Triage Tool... in Papua New Guinea," *The Lancet Regional Health* 13 (2021): 100194.

Outperforming Human Triage

Manual triage relies on stressed, cognitively overloaded nurses who frequently make subjective mathematical errors. Hybrid AI systems drastically outperform humans in rigorous clinical studies:

- **Resource Prediction:** Deep Attention NLP models evaluating unstructured nurse notes achieved an AUC of **88%**, outperforming human triage nurses by **16%** in predicting actual hospital resource utilization.[5]
- **Acuity Accuracy:** Optimized machine learning triage models combining boosting algorithms (like CaB) reach global accuracy levels of **83.3%** across large hospital datasets, eliminating human calculation fatigue.[6]

[5] D. Gligorijevic et al., "Deep Attention Model," *arXiv* (2018).

[6] A. Ahmed et al., "Machine Learning Approach for Developing an E-Triage Tool," *Expert Systems with Applications* (2022).

Conclusion

"By replacing subjective human estimation with a rigorous, three-layered AI architecture, we transform triage from a bottleneck into a high-speed, mathematically guaranteed safety net."

- **NLP Layer:** Understands the patient.
- **Deterministic Layer:** Secures the baseline vitals.
- **Bayesian Layer:** Predicts covert deterioration.

This chatbot does not just digitize a form; it automates clinical reasoning to definitively solve ED overcrowding in resource-constrained environments.